Function of Several Variable

$f : \mathbb{R}^m \to \mathbb{R}^n$ $\qquad y = x$

$f(x_1, x_2, x_3 \ldots x_m) = (f_1, f_2, \ldots f_n)$

$f(x) = x^2$

$f(2) = 4$

e.g $f : \mathbb{R}^3 \to \mathbb{R}^2$

$$f(x_1, x_2, x_3) = [\underbrace{x_1^2 + x_2^2 + x_3}_{f_1}, \underbrace{\sin(x_1 + x_2)}_{f_2}]$$

$$f(1, 2, 0) = [5, \sin(3)]$$

(2) e.g $f : \mathbb{R}^2 \to \mathbb{R}^1$

$$f(x_1, x_2) = (x_1 + x_2)^2$$

Domain of Defination : The Set of Points Where function of Several Variable are going to defined

$f : \mathbb{R}^2 \to \mathbb{R}$ S.t $f(x_1, x_2) = \dfrac{1}{x_1 + x_2}$

$x_1 + x_2 = 0$

$x_1 = -x_2$

$\{(x_1, x_2) \in \mathbb{R}^2, x_1 \neq -x_2\}$

$1/x-1 \qquad \mathbb{R} - \{1\}$

$\{x \in \mathbb{R} ; x \neq 1\}$

Q. $f: \mathbb{R}^2 \to \mathbb{R}^2$ s.t $f(x_1, x_2) = \left[\log x_1, \sin \frac{1}{x_2 - 1}\right]$

Soln: $DOD = \{(x_1, x_2) \in \mathbb{R}^2 \text{ s.t. } x_1 > 0, x_2 \neq 1\}$

Q H.W $f: \mathbb{R}^2 \to \mathbb{R}^3$

$$f(x_1, x_2) = \left(x_1 \cdot \sin \frac{1}{x^2}, \; x_2 \cdot \cos \frac{1}{x_1}, \; x_1 + x_2\right)$$

☆ Limit of a FOSV :-

Let S be any subset of $\mathbb{R}^2$ and (a,b) be a limit point of S f be a function defined on S, then f is send to tend to a limit l as point (x,y) approaches close to (a,b) if $\forall \, \epsilon > 0$

$\exists \; \delta > 0$ s.t $|f(x,y) - l| < \epsilon$ When, ever

$$|x-a| < \delta \;, \; |y-b| < \delta$$

$\lim\limits_{x \to a} f(x) = l$

$f(x) = x^2$

$f(x,y)$

$f(x,y) = yx^2$

$f(1,2) = 2$

$f: \mathbb{R}^m \to \mathbb{R}^n$

$f: \mathbb{R}^2 \to \mathbb{R}$

$$\lim_{(x,y) \to (a,b)} f(x,y) = l$$

e.g. $f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2} & ; (x,y) \neq 0 \\ 0 & ; (x,y) = (0,0) \end{cases}$

Sol^n:- $\lim\limits_{(x,y)\to(0,0)} f(x,y) = ?$

II Path $y = x+2$
$(x,y) \to (0,2)$

Path $y = x$
$\lim\limits_{(x,y)\to(0,0)} \dfrac{x \cdot x}{x^2+x^2} \Rightarrow \lim\limits_{x\to 0} \dfrac{\cancel{x^2}}{2\cancel{x^2}} \Rightarrow \dfrac{1}{2}$

II Path $y = 2x$
$\lim\limits_{x\to 0} \dfrac{x \cdot 2x}{x^2+4x^2} = \dfrac{2\cancel{x^2}}{5\cancel{x^2}} \Rightarrow \dfrac{2}{5}$

$y = mx$
$\lim\limits_{x\to 0} \dfrac{x \cdot (mx)}{x^2+x^2m^2} \Rightarrow \lim\limits_{x\to 0} \dfrac{m\cancel{x^2}}{\cancel{x^2}(1+m^2)} \Rightarrow \dfrac{m}{1+m^2}$

(2) $f(x,y) = \begin{cases} \dfrac{x^2-y^2}{x^2+y^2} & ; (x,y) \neq (0,0) \\ 0 & ; (x,y) = (0,0) \end{cases}$

Sol^n:-
$y = mx$
$\lim\limits_{x\to 0} \dfrac{x^2-(mx)^2}{x^2+m^2x^2} \Rightarrow \dfrac{\cancel{x}}{\cancel{x}}\left(\dfrac{1-m^2}{1+m^2}\right)$

$\Rightarrow \boxed{\dfrac{1-m^2}{1+m^2}}$

(3) $f(x,y) = \begin{cases} \dfrac{x^3+y^2}{x-y} & ; (x,y) \neq (0,0) \\ 0 & ; (x,y) = (0,0) \end{cases}$

Solⁿ:- $y = mx$

$$\frac{x^3+(mx)^2}{x-mx} \Rightarrow \frac{x^3+m^2x^2}{x-mx}$$

$$\frac{\cancel{x^2}(x+m)}{\cancel{x}(1-m)} \Rightarrow \frac{x^2+mx}{1-m} = 0$$

$y = x - mx^3$

$$\lim_{x\to 0} \frac{x^3+(x-mx^3)^2}{x-(x-mx^3)} = \frac{x^3+x^2+m^2x^6-2mx^4}{mx^3}$$

$$\lim_{x\to 0} \frac{1}{m} + \frac{1}{mx} \Rightarrow mx^3 - 2x$$

Does Not exists

Continuity of Several Variable :-

Let f be a function defined on $S \subseteq \mathbb{R}^2$
Let $(a,b) \in S$, then 'f' is said to be

Continuity at (a,b) if

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

Q. $f(x,y) = \begin{cases} \dfrac{x \cdot y}{\sqrt{x^2+y^2}} & ; (x,y) \neq (0,0) \\ 0 & ; (x,y) = (0,0) \end{cases}$

Solⁿ:- $\lim\limits_{x \to 0^-} \dfrac{x \cdot y}{x+y} \Rightarrow \dfrac{0}{0}$

$f: \mathbb{R}^2 \to \mathbb{R}^2$ s.t $f(x,y) = (x^2+3y^2+5,\ x^2+2y+6xy)$
check Continuity at $(3,2)$

That function is always zero and than its a function are Continuous.

✯ Partial Differentiation

$f(x,y) = x^2 + 2xy + 3x^2y^2$

$f'(h) = \lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

$z = f(x,y)$

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x,y)}{h} = f_x$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{\partial z}{\partial y} = \lim_{h \to 0} \frac{f(x, y+h) - f(x,y)}{h} = f_y$$

$$\frac{\partial^2 z}{\partial x^2} = f_{xx}, \quad \frac{\partial^2 z}{\partial y^2} = f_{yy}$$

$$\frac{\partial^2 f}{\partial x \partial y} \Rightarrow \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = f_{xy}$$

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \Rightarrow f_{yx}$$

Q $f(x,y) \Rightarrow x^2y + 2xy$

Soln:- $f_x = \frac{\partial f}{\partial x} = 2xy + 2y$

$f_y = \frac{\partial f}{\partial y} = x^2 + 2x$

$f_{xx} = \frac{\partial^2 f}{\partial x^2} = 2y + 0 \Rightarrow 2y$

$f_{yy} = \frac{\partial^2 f}{\partial y^2} \Rightarrow 0$

$f_{xy} = \frac{\partial^2 f}{\partial x\,\partial y} \Rightarrow \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) \Rightarrow 2x + 2$

$f_{yx} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \Rightarrow 2x + 2$

☆ Homogeneous function :-

$$f(x,y) = a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \dots + a_ny^n$$

$$f(x,y) = x^n\left[a_0 + a_1\left(\frac{y}{x}\right) + \dots + a_n\left(\frac{y}{x}\right)^n\right]$$

$$a_1x^{n-1}y = x^{n-1}\left[a_1 y\right]$$

$$x^n\left[a_1\left(\frac{y}{x}\right)\right] = x^1 \cdot x^{n-1}\left[a_1 \cdot \frac{y}{x}\right]$$

$$= x^n\left[a_1 \cdot \frac{y}{x}\right]$$

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Concept based limits Questions :

(1) $f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2} & ; (x,y) \neq 0 \\ a & ; (x,y) = (0,0) \end{cases}$

Soln : $f(0,0) = a$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$$

- Approach $(0,0)$ along the x-axis $y=0$

$$f(x,y) = f(x,0) = \frac{x(0)}{x^2+0^2} = \frac{0}{x^2} = 0$$

- Approach $(0,0)$ along the line $y=x$

$$f(x,y) \to f(x,x) = \frac{x^2}{x^2+x^2} = \frac{\cancel{x^2}}{2\cancel{x^2}} = \frac{1}{2}$$

$\therefore$ Since $\lim_{(x,y)\to(0,0)} f(x,y) \neq f(0,0)$

$f(x,y)$ is not Continuous at $(0,0)$.

(2) $f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

Ans : $f(0,0) = 0$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}}$$

$x = r\cos\theta$

$y = r\sin\theta$

$r^2 = x^2+y^2$

$$\frac{r^{\cancel{2}}\cos\theta\sin\theta}{\cancel{r}} \Rightarrow \lim_{r\to 0^+} r\cos\theta\sin\theta$$

$$= 0(\cos\theta\sin\theta)$$

$$= 0$$

Since $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0)$

$g(x,y)$ is Continous at $(0,0)$

Euler's Theorem

If $z=f(x,y)$ be a homogeneous fx^n of x and y of degree n,
Then

$$\frac{x\partial f}{\partial x}+y\frac{\partial f}{\partial y}=nf$$

Q If $u=3x^2yz+5xy^2z+4z^4$
Then verify $\frac{x\,du}{dx}+y\frac{du}{\partial y}+\frac{z\,\partial u}{\partial z}=?$

~~$\frac{x\,du}{dx}=x(6x)+y$~~

$$\frac{x\,du}{dx}=6x^2yz+x5y^2z+0$$

$$\frac{y\,du}{dy}=3x^2zy+5x2y^2z+0$$

$$\frac{z\,du}{dz}=3x^2yz+5xy^2z+16z^4$$

$$6x^2yz+5xy^2z+3x^2yz+2.5xy^2z+3x^2yz+5xy^2z+16z^4$$

$$12x^2yz+4.5xy^2z+16z^4$$

$4[3x^2yz + 5xy^2z + 4z^4]$

$\Rightarrow 4u$ Hence proved

Thm → If f is a homogeneous function of degree n Then

(a) $x\frac{d^2f}{dx^2} + y\frac{d^2f}{dxdy} = (n-1)\frac{df}{dx}$ — (1)

(b) $x\frac{d^2f}{dxdy} + y\frac{d^2f}{dy^2} = (n-1)\frac{df}{dy}$ — (2)

(c) $x^2\frac{d^2f}{dx^2} + 2xy\frac{d^2f}{dxdy} + y^2\frac{d^2f}{dy^2} = n(n-1)f$ — (3)

Proof

(1) $x\frac{df}{dx} + y\frac{df}{dy} = n.f$

P.d w.r.t to x

$\frac{df}{dx} + x\frac{d^2f}{dx^2} + y.\frac{d^2f}{dxdy} = n.\frac{df}{dx}$

$\frac{x^2f}{dx^2} + y\frac{d^2f}{dxdy} = n.\frac{df}{dx} - \frac{df}{dx}$

$\boxed{\frac{x^2f}{dx^2} + y\frac{d^2f}{dxdy} \Rightarrow (n-1)\frac{df}{dx}} \rightarrow$ (a)

$$x\frac{df}{dx} + y\frac{df}{dy} = n.f$$

P.d w.r. to y

$$x.\frac{d^2f}{dxdy} + \frac{df}{dy} + y\frac{d^2f}{dy^2} = n.\frac{df}{dy}$$

$$x\frac{d^2f}{dxdy} + y\frac{d^2f}{dy^2} = n.\frac{df}{dy} - \frac{df}{dy}$$

$$\boxed{x\frac{d^2f}{dxdy} + y\frac{d^2f}{dy^2} = (n-1)\frac{df}{dy}} \Rightarrow b$$

$$x\frac{df}{dx} + y\frac{df}{dy} = n.f$$

Diff w.r. to x

$$x\frac{d^2f}{dx^2} + \frac{df}{dx} + y\frac{d^2f}{dxdy} = n.\frac{df}{dx}$$

Multiply by x

$$\frac{x^2d^2f}{dx^2} + \frac{xdf}{dx} + xy\frac{d^2z}{dxdy} = nx.\frac{df}{dx}$$

$$\Rightarrow \frac{x^2d^2f}{dx^2} + xy\frac{d^2f}{dxdy} = (n-1)\,x.\frac{df}{dx}$$

Similarly for y.

$$y^2 \frac{d^2f}{dy^2} + xy \frac{d^2f}{dx\,dy} = (n-1)\, y \frac{df}{dy}$$

Now add the equations

$$x^2\frac{d^2f}{dx^2} + y^2\frac{d^2f}{dy^2} + 2xy\frac{d^2f}{dx\,dy} = (n-1)\left[x\frac{df}{dx} + y\frac{df}{dy}\right]$$

$$\boxed{x^2\frac{d^2f}{dx^2} + 2xy\frac{d^2f}{dx\,dy} + y^2\frac{d^2f}{dy^2} = n(n-1)\, f} \quad -(c)$$

Q $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, then, Prove that $x\frac{du}{dx} + y\frac{du}{dy} = \tan u$

Soln $\sin u = \left(\frac{x^2+y^2}{x+y}\right)$

$z = \left(\frac{x^2+y^2}{x+y}\right)$ where, $z = \sin u$

$= \frac{x^{\cancel{2}}\,(1 + y^2/x^2)}{\cancel{x}\,(1 + y/x)}$ order $= 2-1 = 1$

$\Rightarrow x F(y/x)$

Where, z is a Homogeneous function of degree 1.

$\therefore$ By Euler's theorem

$\therefore$ By $x\frac{dz}{dx} + y\frac{dz}{dy} = z$

$$x\frac{d\sin u}{dx} + y\frac{d(\sin u)}{dy} = n\sin u$$

$$x(\cos u)\frac{du}{dx} + y(\cos u)\frac{du}{dy} = n\sin u$$

$$\cos u\left(x\frac{du}{dx} + \frac{du}{dy}\right) = n\sin u$$

$$\boxed{x\frac{du}{dx} + \frac{du}{dy} = n\tan u}$$

Q $u = \tan^{-1}(y/x)$ Then find $x\frac{du}{dx} + y\frac{du}{dy} = ?$

Soln $\tan u = (y/x)$

$$xF(y/x)$$

when z, is a homogeneous function of degree 1.

$\therefore$ By Euler's theorem

$$x\frac{dz}{dx} + y\frac{dz}{dy} = z$$

$$x\frac{d\tan u}{dx} + y\frac{d\tan u}{dy} = n\tan u$$

$$x\sec^2 u\frac{du}{dx} + y(\sec^2 u)\frac{du}{dy} = n\tan u$$

$$x\frac{du}{dx} + y\frac{du}{dy} = n\frac{\tan u}{\sec^2 u}$$

Total Differential Coefficient

$u = f(x,y)$, Where $x = \phi(t)$, $y = \psi(t)$

$$\frac{du}{dt} = \frac{du}{dx} \cdot \frac{dx}{dt} + \frac{du}{dy} \cdot \frac{dy}{dt}$$

Q If $Z = xy^2 + x^2y$, $x = at^2$, $y = 2at$

Solⁿ: $$\frac{dz}{dt} = \frac{dz}{dx} \cdot \frac{dx}{dt} + \frac{dz}{dy} \cdot \frac{dy}{dt} \quad -(1)$$

$z = xy^2 + x^2y.$

P.D wrt x & y

$$\frac{dz}{dx} = y^2 + (2x)y \quad , \quad \frac{dz}{dy} = x(2y) + x^2$$

$x = at^2$, $y = 2at$

$$\frac{dx}{dt} = 2at \quad , \quad \frac{dy}{dt} = 2a$$

$$\frac{dz}{dt} = (y^2 + 2xy)(2at) + (2xy + x^2)(2a)$$

$$\cancel{\frac{dz}{dt} = 2aty^2 + 4xy.at + 4axy + 2ax^2}$$

$$\cancel{\frac{dz}{dt} = 2aty^2 + 4xyat + 4axy + 2ax^2}$$

$= \left(4a^2t^2 + 2(at^2)(2at)\right)(2at) + (a^2t^4 + 2(at^2)(2at))(2a).$

$= (4a^2t^2 + 4a^2t^3)(2at) + (a^2t^4 + 4a^2t^3)(2a)$

$= 8a^3t^3 + 8a^3t^4 + 2a^3t^4 + 8a^3t^3.$

$= 16a^3t^3 + 10a^3t^4 = 2a^3t^3(8+5t).$

$$\boxed{\frac{dz}{dt} = 2a^3t^3(8+5t).}$$

Q If $u = x^2 - y^2$ Where $x = e^t \cos t$, $y = e^t \sin t$ Then Prove that $\frac{du}{dt} = 2e^{2t}(\cos 2t \cdot \sin 2t)$

$\frac{du}{dt} = \frac{du}{dx} \cdot \frac{dx}{dt} + \frac{du}{dy} \cdot \frac{dy}{dt}$

$\frac{du}{dx} = 2x \,(e^t\cos t - e^t \sin t) + (-2y)[e^t \sin t + e^t \cos t]$

$= 2x \cdot e^t \cos t - 2x e^t \sin t - 2y e^t \sin t - 2y e^t \cos t$

$= 2e^t[x\cos t - x\sin t] - 2e^t[y\sin t + y\cos t]$

$= 2e^t[e^t\cos^2 t - e^t \sin t\cos t] - 2e^t[e^t \sin^2 t + e^t \sin t \cos t]$

$= 2e^{2t}[\cos^2 t - \sin^2 t] - 2e^{2t}\sin 2t$

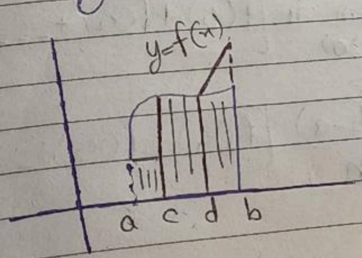
$2e^{2t}[\cos 2t - \sin 2t]$ $\quad [\because \sin 2t = 2\sin t\cos t]$

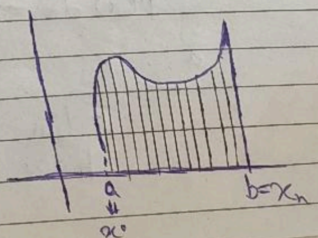
$\Rightarrow$ 1 Baki agea

1. Basic :-

- $\int x^n = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$
- $\int \frac{1}{x} dx = \log_e x + c$
- $\int e^x dx = e^x + c$
- $\int a^x dx = \frac{a^x}{\log_e a} + c$
- $\int \cos x \, dx = \sin x + c$
- $\int \sin x \, dx = -\cos x + c$
- $\int \sec^2 x \, dx = \tan x + c$
- $\int \text{cosec}^2 x \, dx = -\cot x + c$
- $\int \sec x . \tan x \, dx = \sec x + c$
- $\int \text{cosec} x \cot x \, dx = -\text{cosec} x + c$

$y = f(x), \quad [a, b]$





Definite integration

$$a = x_0$$
$$x_1 = x_0 + h$$
$$x_2 = x_0 + 2h$$
$$x_3 = x_0 + 3h$$
$$\vdots$$
$$x_n = x_0 + nh$$
$$b = a + nh$$

$$b - a = nh \qquad \boxed{h = \frac{b-a}{n}}$$

$$r_n = h.f(x_0) + hf(x_1) + \dots + hf(x_{n-1})$$

$$r_n = h\left[f(x_0) + f(x_1) + \dots + f(x_{n-1})\right] \quad -(1)$$

$$R_n = h\left[f(x_1) + f(x_2) + \dots + f(x_n)\right] \quad -(2)$$

$$\text{as } n \to \infty$$
$$r_n = R_n$$

$$\int_a^b f(x)\,dx = r_n = R_n$$

$$\int_a^b f(x)\,dx = \lim_{h \to \infty} h\left[f(x_1) + f(x_2) + \dots + f(x_n)\right]$$

$$= \lim_{h \to \infty} \frac{(b-a)}{n}\left[f(x_1) + f(x_2) + \dots + f(x_n)\right]$$

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$$\Rightarrow \lim_{n\to\infty} \frac{(b-a)}{n} \sum_{i=1}^{n} f(x_i)$$

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(1)

Q $x^3+y^3-3ax^2=0 \quad \frac{dy}{dx}=?, \ \frac{d^2y}{dx^2}=?$

$y^3 = 3ax^2 - x^3$

$3y^2\frac{dy}{dx} = 6ax - 3x^2$

$\frac{dy}{dx} = \frac{\cancel{3}(2ax-x^2)}{\cancel{3}y^2}$

$\boxed{\frac{dy}{dx} = \frac{2ax-x^2}{y^2}}$

$\frac{d^2y}{dx^2} = \frac{y^2(2a-2x)-(2ax-x^2)2y\,dy/dx}{(y^2)^2}$

$= \frac{y^2(2a-2x)-(2ax-x^2)2\cancel{y}\left(\frac{2ax-x^2}{y^{\cancel{2}}}\right)}{y^4}$

$= \frac{y^3(2a-2x)-(2ax-x^2)(4ax-2x^2)}{y^5}$

$\frac{(3ax^2-x^3)(2a-2x)-\{8a^2x^2-4ax^3-4ax^3+2x^4\}}{y^5}$

$= \frac{6a^2x^2-\cancel{2ax^3}-\cancel{6ax^3}+\cancel{2x^4}-8a^2x^2+\cancel{4ax^3}+\cancel{4ax^3}-\cancel{2x^4}}{y^5}$

$\Rightarrow \frac{6a^2x^2-8a^2x^2}{y^5} \Rightarrow \frac{-2a^2x^2}{y^5}$

$\boxed{\frac{d^2y}{dx^2} = \frac{-2a^2x^2}{y^5}}$ $\Rightarrow$ Hence proved

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Jacobian Theorem :-

If U and V are function of two Independent Variable x and y then jacobian of U and V w.r.t x & y is given by

$$\frac{\partial(u,v)}{\partial(x,y)} = J\left[\frac{u,v}{x,y}\right] = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

or

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{du}{dx} & \frac{du}{dy} & \frac{du}{dz} \\ \frac{dv}{dx} & \frac{dv}{dy} & \frac{dv}{dz} \\ \frac{dw}{dx} & \frac{dw}{dy} & \frac{dw}{dz} \end{vmatrix}$$

(Q) if $x = r\cos\theta$ and $y = r\sin\theta$ find $\frac{\partial(x,y)}{\partial(r,\theta)}$

Soln :-

$$= \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$\frac{dx}{dr} = \frac{\partial r\cos\theta}{\partial r} = \cos\theta$

$\frac{dx}{\partial\theta} = -r\sin\theta$

$\frac{dy}{\partial r} = \frac{\partial r\sin\theta}{\partial r} = \sin\theta \; ; \frac{dy}{d\theta} = r\cos\theta$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} \Rightarrow r\cos^2\theta + r\sin^2\theta$$

$$\Rightarrow r\left(\sin^2\theta + \cos^2\theta\right)$$

$$\Rightarrow r \quad \underline{\text{Ans}}$$

<u>Ques</u> If $u_1 = \dfrac{x_2 x_3}{x_1}$, $u_2 = \dfrac{x_3 x_1}{x_2}$, $u_3 = \dfrac{x_1 x_2}{x_3}$

Then prove that $\dfrac{\partial(u_1, u_2, u_3)}{\partial(x_1, x_2, x_3)} \Rightarrow 4$

$$\Rightarrow \begin{vmatrix} \dfrac{du_1}{dx_1} & \dfrac{du_1}{dx_2} & \dfrac{du_1}{dx_3} \\ \dfrac{du_2}{dx_1} & \dfrac{du_2}{dx_2} & \dfrac{du_2}{dx_3} \\ \dfrac{du_3}{dx_1} & \dfrac{du_3}{dx_2} & \dfrac{du_3}{dx_3} \end{vmatrix} = \begin{vmatrix} \dfrac{-x_2x_3}{x_1^2} & \dfrac{x_3}{x_1} & \dfrac{x_2}{x_1} \\ \dfrac{x_3}{x_2} & \dfrac{-x_1x_3}{x_2^2} & \dfrac{x_1}{x_2} \\ \dfrac{x_2}{x_3} & \dfrac{x_1}{x_3} & \dfrac{-x_1x_2}{x_3^2} \end{vmatrix}$$

$$= \begin{vmatrix} \dfrac{-x_2 x_3}{x_1^2} & \dfrac{x_3 . x_1}{x_1^2} & \dfrac{x_2 . x_1}{x_1^2} \\ \dfrac{x^3 . x_2}{x_2^2} & \dfrac{-x_1 x_3}{x_2^2} & \dfrac{x_1 . x_2}{x_2} \\ \dfrac{x_2 . x_3}{x_3^2} & \dfrac{x_1 . x_3}{x_3^2} & \dfrac{-x_1 x_2}{x_3^2} \end{vmatrix}$$

$$= \frac{1}{x_1^2+x_2^2+x_3^2}\begin{vmatrix} -x_2x_3 & x_3.x_1 & x_2.x_1 \\ x_3.x_2 & -x_1x_3 & x_1.x_2 \\ x_2.x_3 & x_1.x_3 & -x_1.x_2 \end{vmatrix}$$

$$\begin{array}{c} \quad\quad + \quad - \quad + \\ = \frac{x_1^2 + x_2^2 + x_3^2}{x_1^2\ x_2^2\ x_3^2}\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} \end{array}$$

$$= -1[1-1] - 1[-1-1] + 1[1+1]$$

$$= 0 + 2 + 2 = 4$$ Hence proved

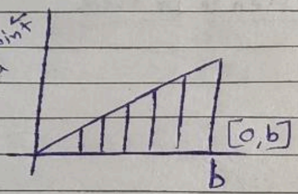
☆ Existence of definate integercation :-

Them :- A Continous function is Integrable. That is If a function f is Continuous on an interval [a, b], then its definite integral Over [a, b] exists.

$$\int_0^b x\,dx = \frac{x^2}{2}\Big|_0^b = \frac{b^2}{2}$$

Reiman's Theorem :-

exam point of you



$$h = \frac{b-a}{n} = \frac{b-0}{n} = \frac{b}{n}$$

$$P = \left\{0, \frac{b}{n}, \frac{2b}{n}, \frac{3b}{n}, \cdots, \frac{nb}{b}\right\}$$

$$\int_0^b x\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\cdot h$$

$$= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{ib}{n}\cdot\frac{b}{n} \Rightarrow \lim_{n\to\infty} \frac{b^2}{n^2} \sum_{i=1}^{n} i$$

$$= \lim_{n\to\infty} \frac{b^2}{n^2}\left[1+2+\cdots+n\right]$$

$$\Rightarrow \lim_{n\to\infty} \frac{b^2}{n^2}\left[\frac{n(n+1)}{2}\right]$$

$$\Rightarrow \lim_{n\to\infty} \frac{b^2}{2\cancel{n^2}} \times \cancel{n^2}\left[1 + \underbrace{\cancel{\frac{1}{n}}}_{0}\right]$$

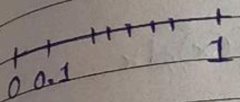
$1/n \Rightarrow$ infinitly because $\lim_{n\to\infty}$

$$\Rightarrow \frac{b^2}{2}$$

Q $\int_0^b x^2\,dx$ By use of Raman's Theorem

Q $\int_0^1 f(x) = \begin{cases} 1, & x \in Q \\ 0, & x \in Q \end{cases}$

$$h = \frac{1}{n}$$



$$L = \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\cdot\frac{1}{n} = \lim_{n\to\infty} \sum 0\cdot\frac{1}{n}$$

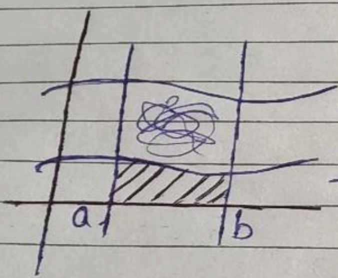
$$U = \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\cdot h = \lim_{n\to\infty} \sum_{i=1}^{n} 1\cdot h$$

$$= 1.$$

$$\star \quad \int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

$$\circ \quad \int_a^b f(x)\,dx = \int_b^c f(x)\,dx = \int_a^b f(x)\,dx$$

$\circ$



$$\int_a^b k f(x)\,dx =$$

$$k \int_a^b f(x)\,dx.$$

Beta and Gamma functions :-

* Beta / Eulers Integer Ist

(1) $B(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1}\, dx \;;\; m,n>0$

$B(6,5) = \int_0^1 x^5 (1-x)^4\, dx$

(2) $B(m,n) = \beta(n,m)$

L.H.S $\quad B(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1}\, dx$

Put $x = 1-y$ — (*)

$\hookrightarrow dx = -dy$

$= -\int_1^0 (1-y)^{m-1} (1-(1-y))^{n-1}\, dy$

$= \int_0^1 y^{n-1} \cdot (1-y)^{m-1}\, dy$

$= B(n,m)$ Hence proved

(2) $B(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\, dx$

L.H.S $\quad B(m,n) = \int_0^1 x^{m-1} (1-x)^{n-1}\, dx$

Put $x = \frac{y}{1+y}$

$$dx = \frac{1}{1+y} \; ; \; dx = \frac{1}{(y+1)^2}\, dy$$

$$\int_0^\infty \left(\frac{y}{1+y}\right)^{m-1} \left(1 - \frac{y}{1+y}\right)^{n-1} \frac{1}{(y+1)^2}\, dy$$

$$= \int_0^\infty \left(\frac{y}{1+y}\right)^{m-1} \left(1 - \frac{y}{1+y}\right)^{n-1} \frac{1}{(y+1)^2}\, dy$$

$$= \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n-2+2}} \times \frac{1}{(y+1)^{n-1}} \times \frac{1}{(y+1)^2}\, dy$$

$$\int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n-2+2}}$$

$$= \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\, dy$$

3) $B(m,n) = 2\int_0^{\pi/2} \sin^{2n-1}\theta \; \cos^{2n-1}\theta \, d\theta$

$$\int_0^1 x^{m-1}(1-x)^{n-1}\, dx$$

$[x = \sin^2\theta$

Gamma function

$$\Gamma n = \int_0^\infty e^{-x} \cdot x^{n-1}\, dx \; ; n>0$$

$$\Gamma 10 = \int_0^\infty e^{-x} \cdot x^{10-1}\, dx$$
$$= \int_0^\infty e^{-x} \cdot x^{9}\, dx$$

(1) $\Gamma(n+1) = n\Gamma n \; ; n>0$

$\Gamma 10 = \Gamma(9+1) \Rightarrow 9\Gamma 9$

(2) $\Gamma n = (n-1)! \; ; n>0$

$\Gamma 2 = 1\Gamma 1 = 1 \times 0! = 1$

(3) $\Gamma 0 = \infty$, $\Gamma(-n) = \infty$, $n \in \mathbb{N}$

Standard Integral functions :-

$$\int_0^\infty x^{n-1} e^{-ax}\, dx = \frac{\Gamma n}{a^n}$$

$$\Gamma n = a^n \int_0^\infty x^{n-1} \cdot e^{-ax}\, dx$$

L.H.S $\Gamma n = \int_0^\infty e^{-x} \cdot x^{n-1}\, dx$

Put $x = az$

$\rightarrow dx = a\, dz$

$$= a\int_0^\infty e^{-az} \cdot (az)^{n-1}\, dz$$

$$= a^n \int_0^\infty e^{-az}\, z^{n-1}\, dz$$

$$\Gamma n = a^n \int_0^\infty e^{-ax}\, x^{n-1}\, dx$$

$$\frac{\Gamma n}{a^n} = \int_0^\infty e^{-ax}\, x^{n-1}\, dx$$

Hence proved

★ **Result :-**

$$B(m,n) = \frac{\Gamma m \, \Gamma n}{\Gamma (m+n)} \qquad (m, n > 0)$$

✳ $\Gamma \frac{1}{2} = \sqrt{\pi}$

eg $\int_0^\infty e^{-x} \cdot x^6 \, dx$

$= \int_0^\infty e^{-x} \, x^{7-1} \, dx \Rightarrow \Gamma 7$

$= 6!$ Ans

$\beta(7,6) = \int_\infty^1 x^6 (1-x)^5 \, dx$

$\Rightarrow \dfrac{\Gamma 7 \, \Gamma 6}{\Gamma 7+6} = \dfrac{6! \times 5!}{12!}$

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$$\int_0^\infty e^{-x} \cdot x^{n-1} dx = \Gamma n$$

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Q Evaluate $\int_0^\infty \sqrt{x} \cdot e^{-x^3} dx$

Sol^n: Put $x^3 = t \rightarrow x = t^{1/3}$

$3x^2 dx = dt$

$dx = \frac{dt}{3x^2}$

$dx = \frac{dt}{3t^{2/3}}$

$\int_0^\infty t^{1/6} \cdot e^{-6} \frac{dt}{3t^{2/3}}$

$\frac{1}{3}\int_0^\infty t^{\frac{1}{6}-\frac{2}{3}} \cdot e^{-t} dt$

$\frac{1}{3}\int_0^\infty t^{\frac{1-4}{6}} \cdot e^{-t} dt$

$\frac{1}{3}\int_0^\infty t^{-1/2} \cdot e^{-t} dt$

$\frac{1}{3}\int_0^\infty e^{-t} \cdot t^{\frac{1}{2}-1} dt$

$\Rightarrow \frac{1}{3} \times \Gamma \frac{1}{2}$

$= \frac{\sqrt{\Gamma 1}}{3}$ Ans

☆ Express in Beta Function:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^5}} dx$$

$$\int_0^1 \frac{x^2}{(1-x^5)^{1/2}} dx \Rightarrow \int_0^1 x^2 (1-x^5)^{-1/2} dx$$

Put $x^5 = t \rightarrow x = t^{1/5}$

$5x^4 dx = dt$

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$dx = \frac{dt}{5x^4} \Rightarrow \frac{dt}{5t^{4/5}}$

$\int_0^1 t^{2/5} (1-t)^{-1/2} \cdot \frac{dt}{5 \cdot t^{4/5}}$

$\frac{1}{5}\int_0^1 t^{2-4/5} (1-t)^{-1/2} \, dt$

$\frac{1}{5}\int_0^1 t^{-2/5} (1-t)^{1/2 - 1} \, dt$

$\frac{1}{5}\int_0^1 t^{3/5 - 1} (1-t)^{1/2 - 1} \, dt$

$\frac{1}{5} B\left(\frac{3}{5}, \frac{1}{2}\right)$

Q $\int_0^2 \sqrt{x} (4-x^2)^{-1/4} \, dx$

Soln:- Put $x^2 = 4t$

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Gamma Formula :-

$$\int_0^{\pi/2} \sin^m\theta \cdot \cos^n\theta \, d\theta = \frac{\Gamma\left(\frac{m+1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{2\,\Gamma\left(\frac{m+n+2}{2}\right)}$$

Q $\int_0^{\pi/2} \sin^3 x \cdot \cos^2 x \, dx$

Soln :- $m=3, \; n=2$

$$\Rightarrow \frac{\Gamma\left(\frac{4}{2}\right)\Gamma\left(\frac{2+1}{2}\right)}{2\,\Gamma\left(\frac{7}{2}\right)} \Rightarrow \frac{1/2 \cdot \sqrt{\pi}}{2 \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}$$

Q $\int_0^{\pi/2} \cos 2x \, dx$

Soln $\int_0^{\pi/2} \cos 2x \, dx = \int_0^{\pi/2} (1 - 2\sin^2 x)\, dx = \frac{\pi}{2} - 2\int_0^{\pi/2} \sin^2 x \, dx$

★ $\Gamma(n+1) = n\,\Gamma(n)$

★ $\Gamma(n) = n!$

★ $\Gamma(1/2) = \sqrt{\pi}$

Q $\Gamma\left(\frac{9}{2}\right) = \Gamma\left(\frac{7}{2}+1\right) \Rightarrow \frac{7}{2}\,\Gamma\left(\frac{7}{2}\right)$

$= \frac{7}{2}\,\Gamma\left(\frac{5}{2}+1\right)$

$= \frac{7}{2} \times \frac{5}{2}\,\Gamma\left(\frac{5}{2}\right)$

$$\frac{7}{2}\times\frac{5}{2}\,\Gamma\left(\frac{3}{2}+1\right) \Rightarrow \frac{7}{2}\times\frac{5}{2}\times\frac{3}{2}\,\Gamma\left(\frac{3}{2}\right) \Rightarrow \frac{7}{2}\times\frac{5}{2}\times\frac{3}{2}\times\Gamma\left(\frac{1}{2}+1\right)$$

$$= \frac{7}{2}\times\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\,\Gamma\left(\frac{1}{2}\right) \Rightarrow \frac{7}{2}\times\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\times\sqrt{\pi}$$

★ $$\boxed{\Gamma n\,\Gamma(1-n) = \frac{\pi}{\sin n\pi}}$$

$$\boxed{B(m,n) = \int_0^{\infty} x^{m-1}(1-x)^{n-1}\,dx = \frac{\Gamma m\,\Gamma n}{\Gamma(m+n)}}$$

Q $\Gamma\left(\frac{-1}{2}\right)$

$n = -\frac{1}{2}$

$$\Gamma\left(\frac{-1}{2}\right)\Gamma\left(\frac{3}{2}\right) = \frac{\pi}{\sin\left(\frac{-\pi}{2}\right)}$$

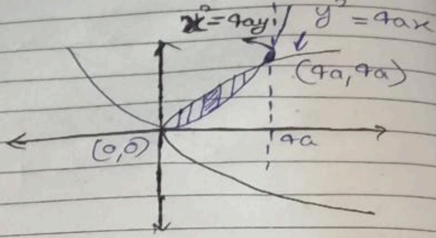
$$\boxed{\Gamma\left(\frac{-1}{2}\right)\Gamma\left(\frac{3}{2}\right) = -\pi}$$

$$\Gamma\left(\frac{-1}{2}\right) = \frac{-\pi}{\frac{1}{2}\times\sqrt{\pi}} \Rightarrow -2\sqrt{\pi}$$ Ans

Q $\Gamma\left(\frac{-3}{2}\right) = ?$; $\Gamma\left(\frac{-5}{2}\right) = ?$

Q: Find the area between the Curve $y^2 = 4ax$ and $x^2 = 4ay$

$$\boxed{y = \frac{x^2}{4a}}$$



$$\left(\frac{x^2}{4a}\right)^2 = 4ax$$

$$\frac{x^4}{16a^2} = \frac{4ax}{1}$$

$$x^4 = 64a^3x$$

$$x^4 - 64a^3x = 0$$

$$x(x^3 - 64a^3) = 0$$

$$x^3 = 64a^3$$

$$\boxed{x = 4a}$$

$$\boxed{y = 4a}$$

$$y = \frac{16a^2}{4a}$$

$$\text{Area} = \int_{x=0}^{4a} \int_{y=\frac{x^2}{4a}}^{2\sqrt{ax}} dx\, dy$$

$$\Rightarrow \int_{x=0}^{4a} \left[\int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy \right] dx = \int_{x=0}^{4a} \left[y \Big|_{\frac{x^2}{4a}}^{2\sqrt{ax}} \right] dx$$

$$\int_0^{4a}\left[2\sqrt{a}\sqrt{x}-\frac{x^2}{4a}\right]dx$$

$$2\sqrt{a}\int_0^{4a}\sqrt{x}\,dx-\frac{1}{4a}\int_0^{4a}x^2\,dx$$

$$2\sqrt{a}\left[\frac{2}{3}x^{3/2}\right]_0^{4a}-\frac{1}{4a}\left[\frac{x^3}{3}\right]_0^{4a}$$

$$4^{3/2}=(2^2)^{3/2}=2^3=8$$

$$\frac{4}{3}\sqrt{a}\,(4a)^{3/2}-\frac{1}{4a}\times\frac{(4a)^3}{3}$$

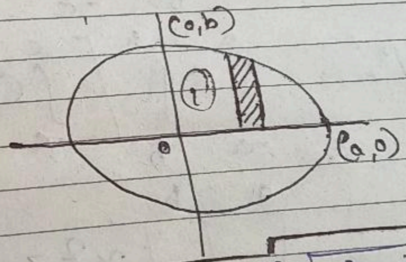
$$\frac{32}{3}a^2-\frac{16a^2}{3}$$

$$\Rightarrow \frac{16a^2}{3}\ \text{Ans}$$

Q Find the Area Covered by the Curve.

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1.$$

$$A=4\int_{x=0}^{a}\int_{y=0}^{\frac{b^2}{a}\sqrt{a^2-x^2}}dx\,dy$$



$$y^2=b^2\left(1-\frac{x^2}{a^2}\right)$$

$$y=b^2\sqrt{\frac{a^2-x^2}{a^2}}$$

$$\boxed{y=\frac{b^2}{a}\sqrt{a^2-x^2}}$$

$$A = 4\int_{x=0}^{a} \left(\frac{b}{a}\sqrt{a^2-x^2}\right)dx$$

$$A = \frac{4b}{a}\int_{x=0}^{a} \sqrt{a^2-x^2}\ dx$$

$$A = \frac{4b}{a}\left[\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_{x=0}^{a}$$

$$A = \frac{4b}{a}\left[\frac{a^2}{2}\frac{\pi}{2}\right] = \pi ab$$

☆ Triple Integration :-

Q $$\int_{-a}^{a}\int_{-b}^{b}\int_{-c}^{c}(x^2+y^2+z^2)\ dx\ dy\ dz$$

Soln :- $$\int_{-a}^{a}\int_{-b}^{b}\left[\int_{-c}^{c}(x^2+y^2+z^2)\ dz\right]dx\ dy$$

$$\int_{-a}^{a}\int_{-b}^{b}\left[x^2z + y^2z + \frac{z^3}{3}\right]_{-c}^{c}dx\ dy$$

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$$\int_{-a}^{a}\int_{-b}^{b}\left\{\left(x^2c+y^2c+\frac{c^3}{3}\right)-\left(-x^2c-y^2c-\frac{c^3}{3}\right)\right\}dx\,dy$$

$$\int_{-a}^{a}\int_{-b}^{b}\left[x^2c+y^2c+\frac{c^3}{3}+x^2c+y^2c+\frac{c^3}{3}\right]dx\,dy$$

$$\int_{-a}^{a}\left[\int_{-b}^{b}\left[2x^2c+2y^2c+2c^3/3\right]dy\right]dx$$

$$\int_{-a}^{a}\left[2x^2cy+\frac{2cy^3}{3}+\frac{2}{3}c^3y\right]_{-b}^{b}dx$$

$$\int_{-a}^{a}\left[\left(2x^2bc+\cancel{2bc}\,\frac{2cb^3}{3}+\frac{2}{3}c^3b\right)-\left(-2x^2bc-\frac{2}{3}cb^3-\frac{2}{3}bc^2\right)\right]dx$$

$$\int_{-a}^{a}\left[4x^2bc+\frac{4c}{3}b^3+\frac{4}{3}bc^3\right]dx$$

$$\left.\frac{4bcx^3}{3}+\frac{4cb^3}{3}x+\frac{4}{3}bc^3x\right|_{-a}^{a}$$

$$\left(\frac{4}{3}bca^3 + \frac{4ab^3c}{3} + \frac{4abc^3}{3}\right) - \left(-\frac{4bca^3}{3} - \frac{4ab^3c}{3} - \frac{4abc^3}{3}\right)$$

$$\frac{8}{3}a^3bc + \frac{8}{3}ab^3c + \frac{8}{3}abc^3$$

$$\frac{8}{3}abc\left[a^2+b^2+c^2\right]$$

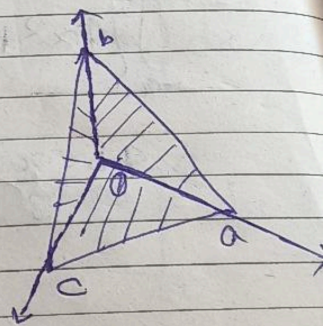
<u>Note</u> :- Area $= \iint dx\,dy$

Volume $= \iiint dx\,dy\,dz$

<u>Q.</u>: Find the Volume Curved by the Curve.?

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

$$V = \iiint_{0} dx\,dy\,dz$$



$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{y}{b} = 1 - \frac{x}{a}$$

$$\boxed{y = \frac{b}{a}(a - x)}$$

$$\frac{z}{c} = 1 - \frac{x}{a} - \frac{y}{b}$$

$$z = c\left[1 - \frac{x}{a} - \frac{y}{b}\right]$$

$$V = \int_0^a \int_0^{b\left(1-\frac{x}{a}\right)} \int_0^{c\left(1-\frac{x}{a}-\frac{y}{b}\right)} dx\,dy\,dz$$

$$V = \int_0^a \left[\int_0^{b(1-x/a)} c\left[1 - \frac{x}{a} - \frac{y}{b}\right] dy\right] dx$$

$$V = \frac{c}{ab} \int_0^a \left[\int_0^{b(1-x/a)} (ab - xb - ya)\,dy\right] d$$

✱ Dirichlet's Integral :

$$\iiint_V x^{l-1} y^{m-1} z^{n-1} \, dx \, dy \, dz = \frac{\Gamma l \, \Gamma m \, \Gamma n}{\Gamma (l+m+n+1)}$$

Where $x \geq 0, y \geq 0, z \geq 0, x+y+z \leq 1$

Q $\iiint_V x^2 y^3 z^2 \, dx \, dy \, dz \; ; \; x \geq 0, y \geq 0, z \geq 0 \; ; \; x+y+z \leq 1$

$$= \frac{\Gamma 3 \, \Gamma 4 \, \Gamma 3}{\Gamma (3+4+3+1)} \Rightarrow$$

✱ Vector Calculus

$\vec{OP} = 2\hat{i} + 3\hat{j} + \hat{k}$